Advanced Programming Practice · 21CSC203P

Unit 3 · Session 1 · SLO 1 — Solved Worksheet

Name: DEEP SAND DEEP SAND | **Reg. No:** RA2511026010557

Unit – 3 : Session – 1 : SLO – 1 | Introduction to Concurrent Programming

Worksheet (Answers)

• Fill: In concurrent programming, multiple THREADS (tasks) can be executed in overlapping time periods.

• T/F: A thread is a lightweight process used in concurrent programming. — TRUE

• MCQ: Commonly used for managing concurrency → A) Semaphore

• Fill: A DEADLOCK condition occurs when two or more threads wait for each other indefinitely.

• T/F: Context switching allows a single processor to execute multiple threads by rapidly switching between them. — TRUE

• MCQ: Mechanism used to protect shared data → A) Mutex

• Fill: RACE CONDITION is the term when two threads access shared data at the same time, leading to unpredictable results.

• T/F: Concurrent programming is only possible on multi-core processors. — FALSE (possible on a single core via context switching)

• MCQ: Example of a synchronization mechanism → A) Condition variable

• Fill: A MUTEX (lock) ensures that only one thread accesses a critical section at a time.